Individual Weekly Report – Week 2

Know Your Data

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Department	Artificial Intelligence and Data Science
Project Title	Youth Smoking and Drug Consumption: From Raw Data to Visual Insights
Role	Data Visualization and Dashboard Development
Week	Week 2 – Know Your Data

Objective of the Week

The aim of this week was to understand the dataset properly. Before creating any chart, it is important to know what data is available, what each variable means and what type of data it is.

Work Done

Analysing the dataset structure: I studied how the dataset is arranged, including the number of rows and columns and the kind of information stored in it. This gave me a basic picture of the size and shape of the data I would be working with.

Studying columns and variables: I went through the available columns one by one and tried to understand what each variable represents. I also noted which variables are numerical and which are categorical, since this decides the type of visualization that can be used.

Identifying suitable visualizations: After understanding the variables, I matched them with suitable charts. For example, histograms and boxplots suit numerical variables, while bar charts suit categories. Scatterplots can be used to compare two numerical variables.

Planning the initial dashboard layout: I made a simple plan of how the dashboard could be arranged, including which sections it might have and where different charts could be placed. This plan was only a first draft and was open to change as the project moved forward.

Tools and Techniques Used

Dataset inspection, variable-type study, chart selection and basic layout planning.

Outcome of the Week

At the end of the week, I had a list of the variables and a set of suitable chart types for each. I also had an initial dashboard layout idea that could guide my work in the coming weeks.

What I Learned

I learned that knowing the data well is the foundation for good visualization. A wrong chart for a variable can give a misleading picture, so understanding data types is very important.

Contribution to the Project

This work helped the team understand the dataset from a visual point of view and gave the project a basic plan for the dashboard.

Challenges Faced

Some variables were not easy to understand at first, and it was difficult to decide the best chart for each of them. I handled this by checking the data type of every variable and by discussing doubtful points with my team members.

Plan for Next Week

In Week 3, I will create initial graphs to check the data quality, including distributions, unusual values, missing values and inconsistencies.

Weekly Summary

Week 2 helped me understand the dataset and connect each variable to a suitable chart. This understanding will guide all the visualizations in the coming weeks.